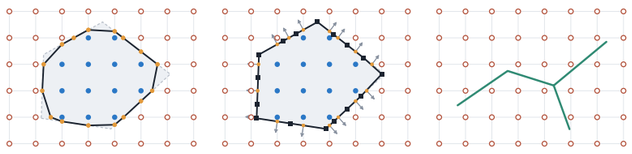
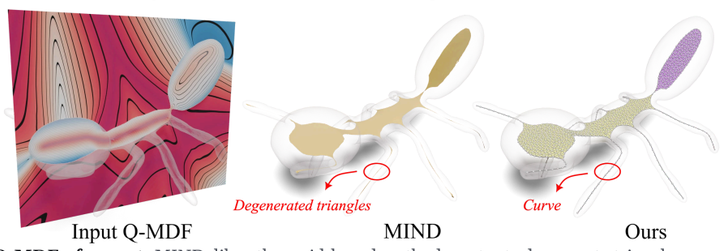
OffsetAxis: UDF Mesh Reconstruction via Offset-Volume Medial Axis Extraction

Qijia Huang · Pierre Kraemer · Dominique Bechmann — ICube, Université de Strasbourg, CNRS — SIGGRAPH Asia 2026 · Kuala Lumpur

Why UDFs are hard to mesh

Marching Cubes uses corner signs to locate edge crossings; Dual Contouring adds normals to keep sharp corners. Both rely on sign changes. A UDF is positive everywhere: crossings cannot be detected by sign changes. Prior work predicts a pseudo-sign (CAP-UDF, MeshUDF, GeoUDF, MIND: Marching Cubes; DualMeshUDF: Dual Contouring). All of these methods depend on the grid resolution: good results need a very fine grid, which makes the meshes extremely dense. Apart from MIND, which outputs degenerate triangles, none of them can output non-manifold structures; Double Covering (DCUDF, DCUDF2) returns two-sheet surfaces there.

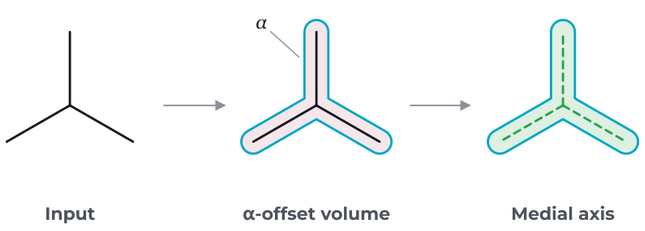




Q-MDF of an ant: MIND, like other grid-based methods, outputs degenerate triangles; OffsetAxis outputs a clean mixed mesh of surfaces and curves.

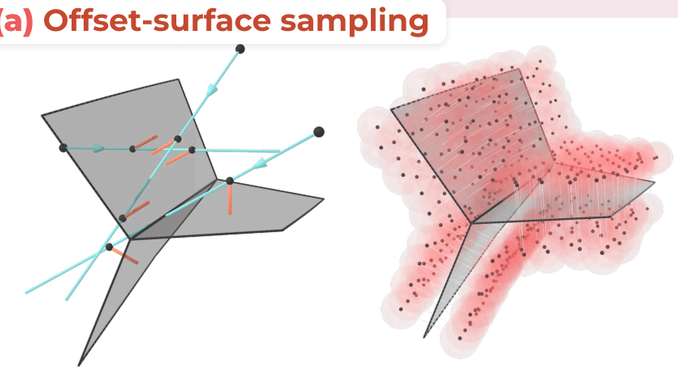
Key idea

Mesh the medial axis of the offset volume.



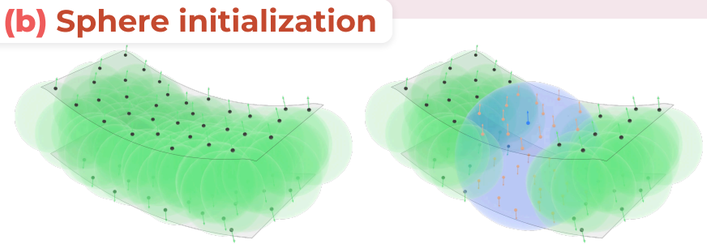
Reconstructing the 0-level set S of a UDF φ is reformulated as extracting the medial axis of its α-offset volume $\Omega_\alpha = \{x \mid \varphi(x) \le \alpha\}$, which is a solid with an inside. If α is smaller than the weak feature size of S, Ω_α and its medial axis are homotopy equivalent to S.

(a) Offset-surface sampling



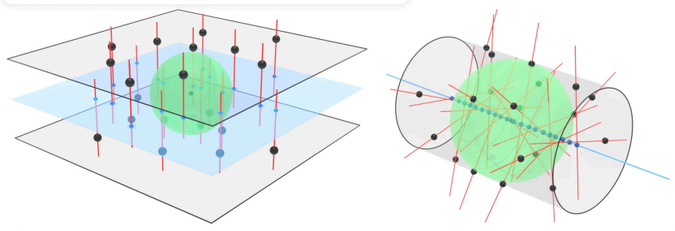
Random rays through an expanded bounding box; sphere tracing finds every crossing with φ = α [Ling et al. 2025], giving seeds on all components. Poisson-disk sampling spreads from the seeds; its radius r must not exceed α. When UDF gradients are unreliable, we fit local normals by PCA on a filtered kNN graph (k = 10).

(b) Sphere initialization



Inspired by Coverage Axis [Dou et al. 2022], we select spheres with a greedy version of it. Each sample gets its medial ball by the shrinking-ball algorithm [Ma et al. 2012]. Coverage: enlarge every radius by δ, take the largest sphere, remove the samples it covers, repeat until all are covered.

(c) Sphere optimization

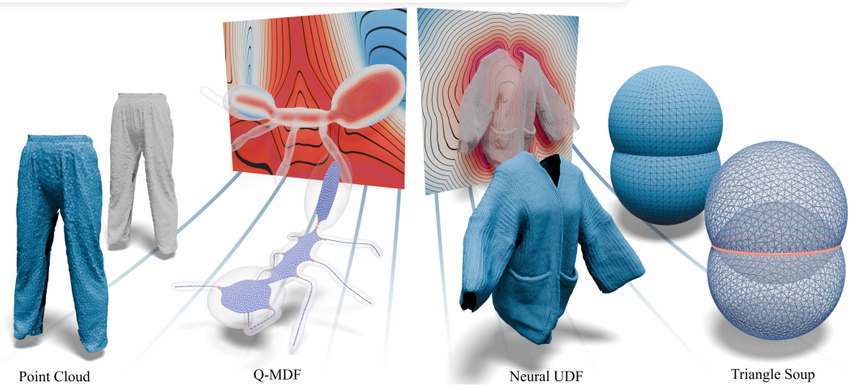


Inspired by VMAS [Huang et al. 2024], we alternate sample-to-sphere clustering and sphere updates. The SQEM Q [Thiery et al. 2013] alone is degenerate on planar and cylindrical parts; the line quadric L [Liu et al. 2025] constrains the centre:

$$E_i = \Sigma_{v_j \in C_i} \left(Q_{v_j}(c_i, r_i) + \mu\, L_{v_j}(c_i) \right), \mu = 0.2$$

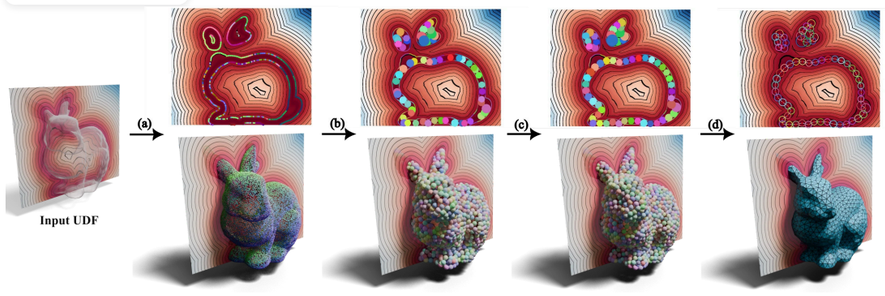
Both terms are quadratic, so each update is one closed-form linear solve.

One general pipeline for diverse inputs



Starting from a function giving the distance to the shape, OffsetAxis extracts structurally coherent meshes from raw point clouds, learned quasi-medial distance fields (Q-MDF), learned UDFs, or triangle soups. The meshes can be open or closed, manifold or non-manifold, and may mix surface and curve parts.

Overview



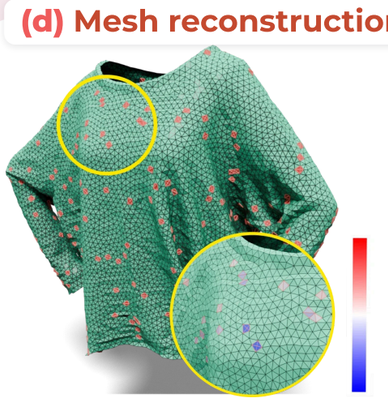
(a) Offset-surface sampling: the α-offset surface is sampled by ray casting.

(b) Sphere initialization: a medial ball is computed for every sample; a coverage then selects a subset.

(c) Sphere optimization: spheres are optimized by variational optimization.

(d) Mesh reconstruction: the mesh connects sphere centres following the connectivity of their clusters.

(d) Mesh reconstruction



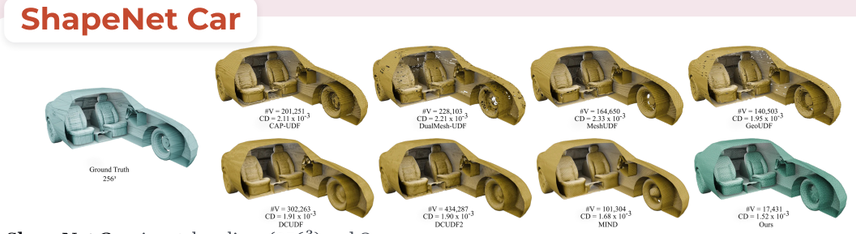
Sphere adjacency gives edges; every 3-clique gives a triangle. 4-cliques create tetrahedra (orange), most of them nearly flat. As a post-processing step, a thinning procedure inspired by MATStruct [Wang et al. 2025] removes simplex pairs (tetrahedron–face, face–edge) in decreasing order of their integrated UDF score (zoom: face colours): simple pairs first; when none is left, the non-simple pair with the largest score, which may expose new simple pairs. It stops when no deletion remains. The output has only triangles and segments.

Open surfaces



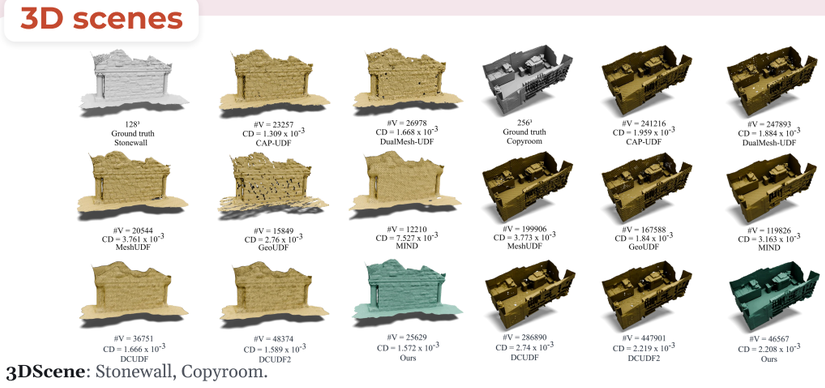
DeepFashion garments: input, baselines and Ours at three resolutions.

ShapeNet Car



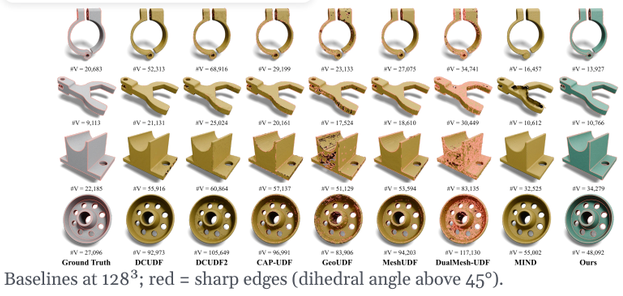
ShapeNet Car: input, baselines (256^3) and Ours.

3D scenes



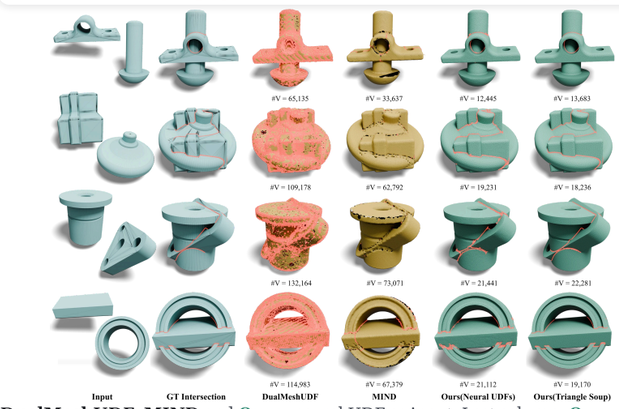
3DScene: Stonewall, Copyroom.

Sharp CAD edges



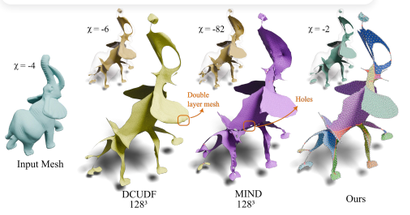
Baselines at 128^3; red = sharp edges (dihedral angle above 45°).

Intersecting parts: non-manifold edges



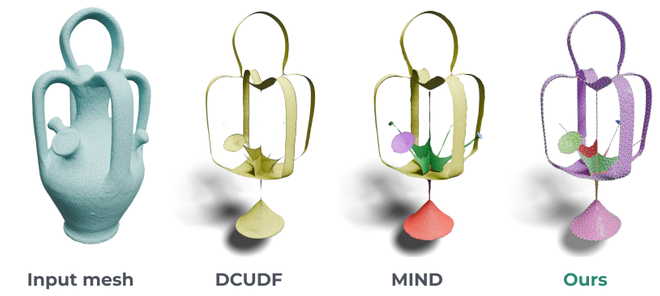
DualMeshUDF, **MIND** and **Ours**: neural UDF as input. Last column: **Ours** with a triangle soup as input.

Non-manifold Q-MDF

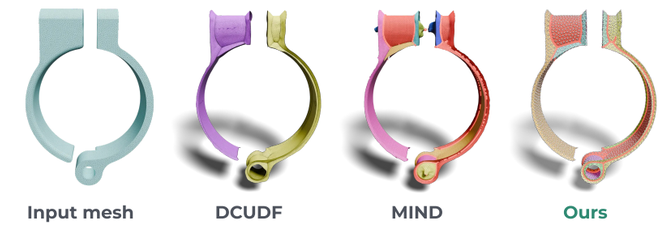


DCUDF produces one duplicated two-layer surface; **MIND** leaves holes with insufficient resolution; **OffsetAxis** recovers the non-manifold edges (red) and splits the mesh into coherent manifold patches (colours).

Teapot · Q-MDF 256^3



ABC_020 · Q-MDF 256^3

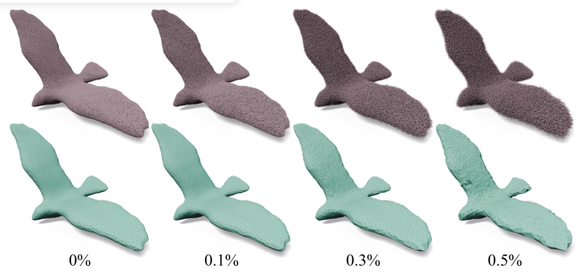


Direct inputs



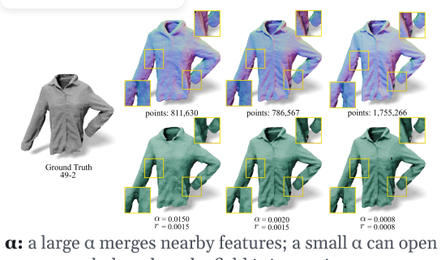
Point clouds and triangle soups can be meshed directly: OffsetAxis only needs to query the unsigned distance to the input. Distance queries are accelerated with a BVH for triangle soups and a kd-tree for point clouds.

Robust to noise

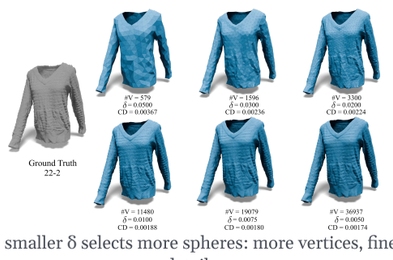


A UDF trained on increasingly noisy point clouds still gives a reasonable mesh that keeps the structure.

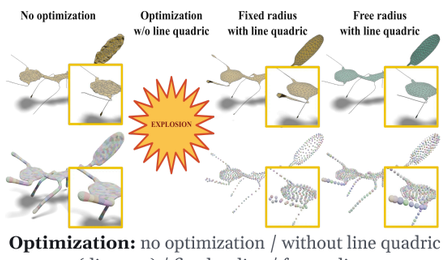
Ablations



α: a large α merges nearby features; a small α can open holes when the field is imprecise.

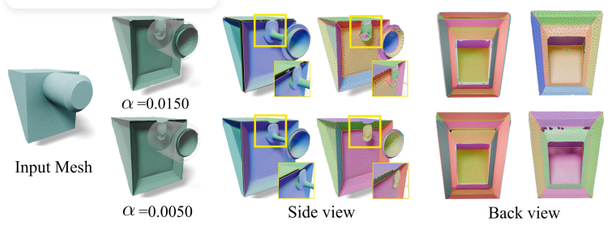


δ: smaller δ selects more spheres: more vertices, finer details.



Optimization: no optimization / without line quadric (diverges) / fixed radius / free radius.

Limitations



No formal topological guarantee: connectivity comes from sphere-cluster adjacency, which is a heuristic. Sometimes no single α fits (learned Q-MDF of a CAD model: large α merges features, small α opens holes).

Project page · 3D demo
huang46u.github.io/OffsetAxis-Page

Talk slides
offsetaxis.pages.dev

Paper
doi:10.1145/3829340.3842256

Acknowledgements. Grand Est Region, grant 22-DOC-067 (Def-RV); French ANR (PIA), grant ANR-22-CE38-0010-01 (POSTURES). Thanks to Dominique Attali.

ICube · Université de Strasbourg · CNRS